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The accuracy of Apple Watch measurements: a living systematic review and meta-analysis



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Apple Watch provides continuous monitoring of physiological and behavioural health metrics, increasingly used to support health-care delivery. Yet, evidence regarding its measurement accuracy remains limited. We aimed to assess the accuracy of measurements from Apple Watch. We searched nine databases from inception to September 24, 2025, with no restrictions on language or publication type. Eligible studies validated any Apple Watch health metric against a criterion method. The primary outcome was the agreement between Apple Watch and the criterion. We included 82 studies, which assessed 14 health metrics (430,052 participants; pooled mean age 41.3 years [SD 13.3]). Bland-Altman meta-analysis showed a small underestimation of heart rate, although limits of agreement (LoA) indicated moderate measurement variability (mean bias -0.27 bpm [95% CI -0.72–0.17]; LoA -7.19 to 6.64). For atrial fibrillation detection, Apple Watch was more specific than sensitive (specificity 0.91 [95% CI 0.81–0.96]; sensitivity 0.79 [95% CI 0.61–0.90]). For blood oxygen saturation, there was low mean bias (-0.04% [95% CI -0.42–0.35]) but wide limits of agreement (-4.00 to 3.94). Accuracy for sleep and step count was moderate, whereas error for energy expenditure was inconsistent and frequently large. Measurement accuracy varied by metric, measurement conditions, and individual physiology. Longitudinal validation of key clinical metrics, including vital signs, is needed to inform clinical practice and policy. This study was registered with PROSPERO, CRD42023481841.

Wearable devices provide personal health monitoring and their clinical role in supporting health-care delivery is growing swiftly¹. They have enabled longitudinal assessment of physiology at scale due to their measurement of health metrics such as heart rate, blood oxygen saturation, and cardiorespiratory fitness^{2,3}. This has allowed early detection of respiratory illness, prediction of cardiovascular risk, and population-level assessment of physical activity^{2,4,5}. Given the current emphasis on personalised medicine and digital phenotyping, there is a growing need for accurate consumer devices that enable the remote capture of digital biomarkers and biometrics⁶.

Compared with traditional methods, wearable devices offer continuous measurement that may facilitate identification of trends in health status and preventative care^{7,8}. Yet, without validation, wearable device measurements may misguide assessment and treatment, potentially resulting in misrepresentations of health or delayed interventions.

Apple Watch (*Apple Inc., California*) is the most widely owned wearable device worldwide, with over 100 million users⁶, and measures several

health metrics that have been associated with cardiovascular and all-cause mortality when assessed using criterion methods^{8–17}. However, its measurement accuracy is not well-established. Existing literature indicates that accuracy is dependent on the individual metric, as well as the measurement conditions¹⁸. Previously, Apple Watch heart rate measurements have shown strong agreement with criterion measures, but factors such as exercise intensity, movement pattern, and skin contact affect accuracy^{19–22}. Conversely, energy expenditure estimates have demonstrated low levels of agreement^{22–24}, and sensitivity and specificity for atrial fibrillation detection range widely between studies²⁵.

This heterogeneity permeates the current literature. Variation in study protocols and criterion methods render comparative analysis of validation studies challenging. Prior systematic reviews and meta-analyses have included a small number of studies, many of which validated Apple Watch software and hardware that has since been discontinued^{23,24,26}. Over the past five years, there has been a substantial increase in validation research,

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however, a contemporary literature synthesis including all health metrics has not been conducted. The yearly update cycle of Apple Watch, and swift advances in machine learning algorithms which underpin measurements, accentuate this issue²⁷.

A continuously updated synthesis of Apple Watch metrics is required. To address this, our review was designed as a living study to provide an up-to-date evaluation of the device's measurement accuracy, in accordance with the *analytical validation* component of the V3 framework²⁸. We defined *health metrics* as any health-related physiological, behavioural, or environmental metric measured natively by Apple Watch. Our aim was to better understand the competencies and boundaries of Apple Watch in clinical and personal health contexts. Our objectives in this systematic review and meta-analysis were to: (1) identify all Apple Watch health metrics that have been validated in primary research studies, (2) evaluate the measurement accuracy of each metric, and (3) identify gaps in the current research.

Results

Following the removal of duplicates, 1202 records were identified. After title and abstract screening, 221 full texts were assessed for eligibility (PRISMA flow diagram, Fig. 1). Articles excluded following full-text review are listed in the Supplementary Information (pp. 18–32). Overall, 82 studies (430,052 participants) were included in this systematic review. Additional results, which include synthesis of hypertension notification, heart rate variability,

sound exposure, and Six-Minute Walk Test distance estimation, along with funnel plots, are provided in Supplementary Note 1.

Fourteen health metrics from all Apple Watch models through to Series 9 and Ultra 2 were validated. Fifty-seven percent of all participants were male, and the median sample size was 44. Information on total sample size was available for 81 of all 82 studies, and male–female split was available for 75 studies. Heart rate was the most frequently validated metric (38 studies), whereas only one study assessed hypertension notification, sound exposure, and heart rate variability. Study characteristics, including criterion methods and sample demographic, are listed in Table 1.

Risk of bias

Overall, 13 (14%) studies were classified as ‘low’ risk of bias, 29 (30%) as ‘some concerns’, and 53 (56%) as ‘high’. Domain 1 (participants) and Domain 4 (statistical analysis) were most frequently rated as high risk. Twenty-six (27%) studies did not appropriately select participants to represent the target population (Domain 1), and 20 (21%) used inappropriate statistical analysis. This included complete exclusion of unsuccessful measurements, use of unsuitable statistical measures of agreement (e.g., *t*-tests), inadequate reporting of missing data, or failure to account for repeated measures. By contrast, Domain 3 (reference standard) was predominantly rated as low risk (85/95 [89%]). Validation protocols, criterion methods, and time intervals between assessments were mostly appropriate.

Fig. 1 | PRISMA flow diagram.

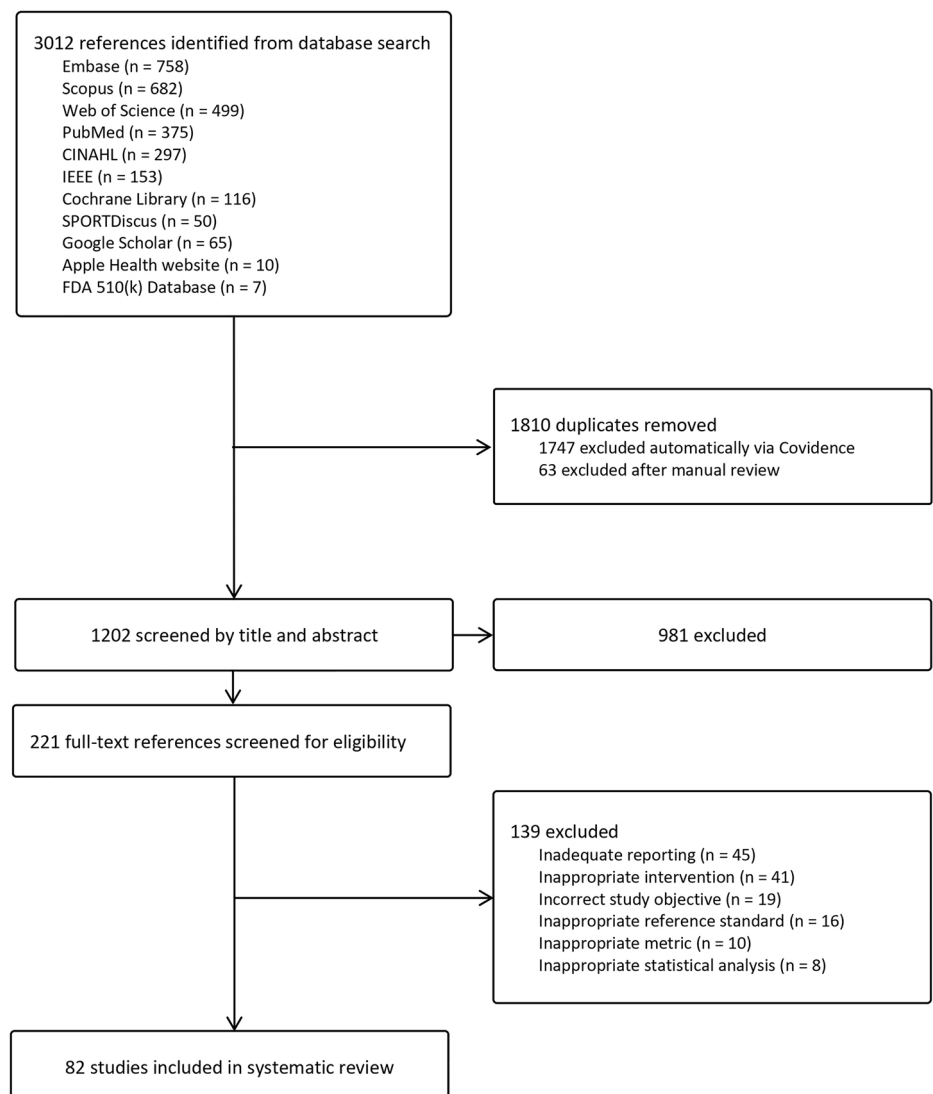


Table 1 | Characteristics of included studies

Primary Author (Year)	Participants (n male); Mean age (SD)	Population	Apple Watch model	Criterion measure	Protocol	Included free-living validation
Heart rate (38 studies)						
O'Grady et al. ⁴⁸	39 (17); 24.6 years (8.2)	Healthy adults	Series 9 and Ultra 2	Polar H10	Resting heart rate measured directly after waking each morning for 14 days.	Yes
Mulholland et al. ⁴³	28 (15); 25 years (5)	Healthy adults, across the range of Fitzpatrick skin types	Series 8	Polar H10	Heart rate at rate, during cycling, and during recovery phase, averaged into 30 s epochs.	No
Khushhal et al. ⁴⁶	112 (112); 50 years (11)	Patients with non-communicable diseases	Series 8	Polar H10	Heart rate at rest and during stationary cycling at 50–70% heart rate reserve.	No
Khushhal et al. ⁴²	260 (260); 47 years (17)	Cardiac patients	Series 8	12-lead ECG	Simultaneous criterion and Apple Watch ECG recordings at rest.	No
Helmer et al. ⁴⁵	10 (3); 63 years [median]	Post-operative patients of abdominal, urological, and non-cardiac thoracic surgery.	Series 7	3-Lead ECG	At rest during hospital stay.	No
Kim et al. ²¹	44 (36); 60.9 years (NR)	Patients with cardiovascular disease	Series 7	12-lead ECG	Treadmill cardiopulmonary exercise test, during exercise and rest phases.	No
Montalvo et al. ¹³²	43 (33); 24.5 years (2.2)	Active and athletic adults	Series 6 and 7	12-lead ECG	Graded treadmill protocol – resting, walking, jogging, running.	No
Støve et al. ²⁹	29 (13); 24.5 years (4)	Healthy adults	Series 6	Polar H10	Barbell and dumbbell resistance exercises, cycle ergometer.	No
Alfonso et al. ¹³³	20 (10); 22 years (2.5)	University students	Series 6	Biopac MP36 ECG	At rest, and low intensity exercise.	No
Ho et al. ³⁴	30 (15); 29.3 years (3.6)	Healthy adults	Series 6	12-lead ECG	Ramp incremental exercise test (cycle ergometer).	No
Uphill et al. ³⁹	50 (50); 30.9 years (4.6)	Male soldiers	Series 5	Polar Team2 Pro	Close quarter combat training.	No
Giggins et al. ³³	10 (5); 30.4 years (8)	Healthy adults	Series 5	Polar H10	Graded treadmill protocol at three (5 min) stages increasing in intensity.	No
Behzadi et al. ⁶⁵	100 (62); 63.7 years (14)	Patients with non-communicable diseases	Series 4	12-lead ECG	Resting heart rate.	No
Bent et al. ¹⁸	53 (21); 25.6 (NR)	Not specified	Series 4	ECG (Bittium Faros 180)	Rest, low intensity exercise, daily activities.	No
Düking et al. ¹³⁴	25 (11); 26 years (7)	Healthy adults	Series 4	Polar H7	Sitting, walking, running at various speeds.	No
Rhee et al. ³⁰	23 (13); 23.6 years (3.2)	Physically active adults	Series 4	Polar H7	Range of activity levels from rest to vigorous intensity exercise.	Yes
Lee et al. ¹³⁵	200 (118); 65.6 years (14.6)	Patients with cardiovascular disease	Series 4	12-lead ECG	Resting heart rate in sitting.	No
Thomas et al. ³⁸	20 (10); NR	Healthy adults	Series 4	Polar H10	Sedentary, walking, running on treadmill.	No
Seshadri et al. ⁶³	50 (36); 61.4 years (10.4)	Patients with cardiovascular disease	Series 4	6-lead telemetry	Resting heart rate.	No
Saghir et al. ⁴⁹	43 (15); 31 years (8.5)	Healthy adults	Series 4	12-lead ECG	Resting heart rate in supine.	No
Al-Kaisey et al. ¹³⁶	20 (NR); 68 years (12)	Patients with AFib and those in sinus rhythm	Series 3	Holter monitor	24-hour monitoring, free-living activities.	Yes
Pasadyan et al. ¹³⁷	50 (34); 29.5 years (9.3)	Healthy athletic adults	Series 3	3-lead ECG	Running at various speeds – graded treadmill protocol.	No
Nelson et al. ³⁵	1 (1); 29 years	Healthy adults	Series 3	3-lead ECG	Free-living: sitting, walking, treadmill running, ADLs, sleeping.	Yes
Bai et al. ³¹	48 (17); 26.8 years (3)	Healthy adults	Series 2	Polar H7	24-hour free living: sedentary, light and moderate physical activity.	Yes
Støve et al. ³⁷	30 (15); 24.8 years (6.3)	Healthy adults	Series 2	Polar H10	Graded exercise protocols on a cycle ergometer and treadmill, plus, rapid arm movements.	No

Table 1 (continued) | Characteristics of included studies

Primary Author (Year)	Participants (n male); Mean age (SD)	Population	Apple Watch model	Criterion measure	Protocol	Included free-living validation
Hwang et al. ¹³⁸	51 (27); 44.4 years (16.6)	Patients with cardiovascular disease	Series 2	12-lead ECG	At rest, and during induced supraventricular tachyarrhythmia.	No
Koshy et al. ⁴⁷	102 (66); 68 years (15)	Patients with cardiovascular disease	Series 1	ECG	Resting heart rate in supine.	No
Nuss et al. ³⁶	30 (15); 22.5 years (3.7)	Young healthy adults	Series 1	12-lead ECG	Graded treadmill protocol.	No
Heyken et al. ¹³⁹	35 (NR); 69.6 (NR)	Patients with cardiovascular disease	Series 1	ECG (number of leads not reported)	Exercise protocol on a cycle ergometer.	No
Huynh et al. ¹⁴⁰	20 (17); 66 years (6.5)	Patients with sleep apnoea in AFib	1 st generation	CARESCAPE Monitor B650	Measured during sleep.	No
Abt et al. ⁴⁰	15 (8); 32 years (10)	Recreationally active adults	1 st generation	Polar T31	Incremental maximal oxygen uptake test on a treadmill.	No
Khushf et al. ¹⁴¹	21 (21); 31.4 years (7.2)	Healthy adults	1 st generation	Polar S810i monitor	Incremental exercise protocol on treadmill.	No
Falter et al. ³²	40 (32); 61.9 years (15.2)	Patients with cardiovascular disease	1 st generation	12-lead ECG	Graded cardiopulmonary exercise test on a cycle ergometer.	No
Eitwy et al. ⁴¹	80 (65); 62 years (13)	Patients with cardiovascular disease	NR	ECG (Quinton Q-Tel RMS)	Treadmill and stationary cycling at steady state exercise at 50-70% HR reserve.	No
Thomson et al. ¹⁴²	30 (15); 23.5 years (3)	Healthy adults	NR	12-lead ECG	Graded treadmill protocol, from <20% to >85% heart rate reserve.	No
Gillinov et al. ¹⁴³	50 (23); 38 years (12)	Healthy adults	NR	Quinton Q-tel RMS ECG	HR was measured at different intensities while on a treadmill, stationary bike, and on an elliptical – both with and without arm levers.	No
Sequeira et al. ¹⁴⁴	24 (11); 53 years (16.4)	Patients with cardiovascular disease	NR	12-lead ECG	During atrial and ventricular stimulation.	No
Wallen et al. ⁴⁴	22 (11); 24 years (5.6)	Healthy adults	NR	3-lead ECG	At rest, and graded exercise tests on a treadmill and cycle ergometer.	No
AFib detection (17 studies)						
Brosia E Gala et al. ⁵⁷	483 (344); 66 years (NR)	Patients with cardiovascular disease	Series 6	12-lead ECG, 2 cardiologists	Simultaneous recordings at rest.	No
Mannhart et al. ⁵³	200 (139); 67 years (IQR 58-75)	Patients with cardiovascular disease	Series 6	12-lead ECG, 2 cardiologists	Consecutive recordings at rest, criterion measurement first.	No
Mannhart et al. ⁵²	117 (85); 65 years (IQR 56-74)	Patients with cardiovascular disease	Series 6	12-lead ECG, 2 cardiologists	Consecutive recordings at rest, criterion measurement first.	No
Peppinkhuizen et al. ⁵⁸	74 (59); 67.1 years (12.3)	Patients with cardiovascular disease	Series 6	12-lead ECG, 2 cardiologists	Consecutive recordings at rest, criterion measurement first.	No
Muller et al. ⁵⁸	93 (73); 68 years (9.9)	Patients with cardiovascular disease	Series 5	6-lead telemetry, 12-lead ECG, 1 blinded physician	Simultaneous readings 3 times daily in hospital setting.	No
Velraeds et al. ⁵⁵	144 (NR); Age NR	Patients with and without cardiovascular disease	Series 5	12-lead ECG, 1 electrophysiologist	Consecutive recordings at rest, criterion measurement first.	No
Wasserlauf et al. ⁵⁰	30 (18); 65.4 years (12.2)	Patients with cardiovascular disease	Series 5	Insertable cardiac monitor (ICM) or cardiac implanted electronic device (CIED)	14-h monitoring for several days. Irregular Rhythm Notifications compared with ICM/CIED.	Yes
Abu-Alrub et al. ⁵¹	200 (112); 62 years (7)	Patients with AFib	Series 5	12-Lead ECG + 2 blinded cardiac electrophysiologists	Consecutive recordings at rest, criterion measurement first.	No
Racine et al. ⁶⁰	734 (426); 66 years (NR)	Patients in sinus rhythm and in arrhythmia	Series 5	12-Lead ECG + 2 blinded cardiac electrophysiologists	Consecutive recordings at rest, criterion measurement first.	No
Scholten et al. ⁵⁴	220 (143); 70 years (10)	Patients with cardiovascular disease	Series 5	12-lead ECG, 1 cardiologists	Consecutive recordings at rest, criterion measurement first.	No
Ford et al. ⁶²	125 (62); 76 years (7)	Patients with cardiovascular disease	Series 4	12-lead ECG, 2 cardiologists	Consecutive recordings at rest, criterion measurement first.	No

Table 1 (continued) | Characteristics of included studies

AFib detection (17 studies)						
Lee et al. ¹³⁵	200 (118); 65.6 years (14.6)	Patients with cardiovascular disease	Series 4	12-lead ECG, 1 cardiologist	Consecutive recordings at rest, criterion measurement first.	No
Saghir et al. ⁶¹	18 (16); Age NR	Patients with cardiovascular disease	Series 4	3-lead ECG, 1 electrophysiologist	Simultaneous measurement.	No
Seshadri et al. ⁶³	50 (36); 61.4 years (10.4)	Patients with cardiovascular disease	Series 4	6-lead telemetry interpreted by 1 blinded cardiologist	Simultaneous measurement at rest 3 times daily for 2 days.	No
Perez et al. ²⁵	419,297; 41 years (13)	Patients with and without cardiovascular disease	Series 1 to 4	ePatch (BioTelemetry, Philips) and 2 blinded clinicians	Free-living monitoring with Apple Watch and ECG patch.	Yes
Wouters et al. ⁶⁴	122 (78); 69 years (IQR 61–77)	Patients with cardiovascular disease	NR	12-lead ECG and 2 blinded cardiologists	At rest, consecutive measurements.	No
Apple ⁶⁶	546 (NR)	Individuals with and without history of AFib	NR	12-lead ECG and 2 blinded cardiologists	Simultaneous recordings with Apple Watch and 12-lead ECG at rest.	No
Blood oxygen saturation (10 studies)						
Walzel et al. ⁷⁶	18 (14); 23.2 years (1.8)	Healthy adults	Series 8	Pulse oximetry (Masimo Radical-7)	Simultaneous readings during induced hypoxemia.	No
Khushf et al. ⁴²	260 (260); 47 years (15)	Patients with disease of metabolic and circulatory system	Series 8	Pulse oximetry (CMS50DL, Contec Medical Systems)	At rest, and whilst stationary, following moderate intensity exercise.	No
Helmer et al. ⁴⁵	10 (3); 63 years [median] (NR)	Post-operative patients, including non-cardiac thoracic surgery.	Series 7	Pulse oximetry (FAST Sensor M1191B, Philips)	Simultaneous measurement during hospital stay. Multiple Apple Watch measurements per day.	No
Rajakarar et al. ⁷²	200 (108); 66 years (18)	Patients with COVID-19	Series 7	Pulse oximetry (Welch Allyn Connex Spot Monitor)	Consecutive readings: criterion then Apple Watch.	No
Jiang et al. ⁷⁷	9 (5); 25 years [median] (range 19–28)	Healthy adults	Series 7	Arterial blood gas	Simultaneous readings during induced hypoxemia.	No
Jiang et al. ⁷¹	49 (31); 64 years [median] (range 38–76)	Patients with disease of respiratory and circulatory system	Series 7	Pulse oximetry (Masimo MightySat Rx)	Simultaneous reading in sitting.	No
Apple ⁴⁵	24 (12); 26.6 years (range 19–40)	Healthy adults	Series 6	Arterial blood gas	Simultaneous readings, including during induced hypoxaemia.	No
Arslan et al. ⁷⁰	167 (99); 70.9 years (11.5)	Patients with disease of respiratory system	Series 6	Arterial blood gas	Simultaneous readings at rest whilst seated.	No
Rafli et al. ⁷⁵	24 (19); 24 years (2)	Healthy adults	Series 6	Pulse oximetry (Masimo Radical-7)	Simultaneous measurement during induced hypoxaemia.	No
Spaccarotella et al. ⁷³	257 (168); 43.2 years (14.3)	Healthy adults and patients with cardiovascular or respiratory disease	Series 6	Pulse oximetry (Neilor Portable, PM10N)	Consecutive readings in sitting. Two measurements per participant.	No
Energy expenditure (8 studies)						
Sun et al. ⁷⁹	11 (1); 22.5 years (1.8)	Physically active adults	Series 6	Indirect calorimetry (Metamax 3B)	Treadmill and ground running at multiple speeds.	No
Uphill et al. ³⁹	50 (50); 30.9 years (4.6)	Male soldiers	Series 5	Indirect calorimetry (Metamax 3B)	Close quarters combat training activity.	No
Duking et al. ¹³⁴	25 (11); 26 years (7)	Healthy adults	Series 4	Indirect calorimetry (Metamax 3B)	Sitting, walking, running at various speeds.	No
Mihyun Lee et al. ⁷⁸	78 (40); 37.1 years (10.8)	Healthy adults	Series 2	Indirect calorimetry (Cosmed K4b2)	Swimming at various speeds.	No
Falter et al. ³²	40 (32); 61.9 years (15.2)	Patients with cardiovascular disease	Sport (1st generation)	Indirect calorimetry (Jaeger Oxycon)	Graded cardiopulmonary exercise test on a cycle ergometer.	No
Nuss et al. ³⁶	30 (15); 22.5 years (3.7)	Healthy adults	Series 1	Indirect calorimetry (ParvoMedics TrueOne 2400)	Graded treadmill protocol, and whilst sitting and standing.	No
Nuss et al. ³⁶	30 (15); 23.5 years (3)	Healthy adults	NR	Indirect calorimetry (ParvoMedics TrueOne 2400)	Graded treadmill protocol, and whilst sitting and standing.	No
Wallen et al. ⁴⁴	22 (11); 24 years (5.6)	Healthy adults	NR	Indirect calorimetry (Metamax 3B)	At rest, and during graded treadmill and cycling protocols.	No

Table 1 (continued) | Characteristics of included studies

ECG waveform (7 studies)						
ECG waveform (7 studies)						
Khushhal et al. ⁴⁶	112 (112); 50 years (112)	Patients with disease of metabolic and circulatory system	Series 8	12-lead ECG	Simultaneous recording with 12-lead ECG and Apple Watch at rest.	No
Buelga Suárez et al. ⁶⁶	26 (18); 75.3 years (18)	Patients with disease of circulatory system	Series 7	12-lead ECG	Simultaneous 12-lead ECG and Apple Watch 30 s recordings at rest.	No
Harmon et al. ⁶⁷	74 (41); 59.2 (NR)	Patients with disease of circulatory system	Series 5	12-lead ECG	30 s Apple Watch reading during a 5 min 12-lead ECG measurement.	No
Klier et al. ⁶⁸	81 (58); 24.9 years (8.6)	Healthy adults	Series 4	12-lead ECG	Simultaneous readings at rest.	No
Behzadi et al. ⁶⁵	100 (62); 63.7 years (14)	Patients with disease of respiratory and circulatory system, and metabolic disease.	Series 4	12-lead ECG, cardiologist	Apple Watch ECG measurement taken immediately after 12-lead ECG measurement.	No
Saghir et al. ⁴⁹	43 (15); 31 years (8.5)	Healthy adults	Series 4	12-lead ECG, analysed by 3 medical residents	Consecutive measurements.	No
Strik et al. ⁶⁹	100 (59); 67 years (7)	Healthy adults and patients with disease of circulatory system	Series 4	12-lead ECG	Consecutive measurements, 12-lead ECG measurement first.	No
Wheelchair push count (3 studies)						
Benning et al. ⁸²	15 (12); 45 years (15.8)	Individuals with spinal cord injuries and paralysis	Series 4	Manual counting	188-meter indoor test circuit.	No
Glasheen et al. ⁸³	30 (18); 47 years (12)	Wheelchair and non-wheelchair users.	Series 1	Manual counting	Wheelchair treadmill and an arm cycle ergometer, and overground obstacle course.	No
Karinharju et al. ⁸⁴	26 (20); 42 years (13)	Active wheelchair users	Series 1	Manual counting	Indoor obstacle course.	No
Step count (3 studies)						
Bunn et al. ⁸¹	20 (10); 26.6 years (11.5)	Adults at risk of cardiovascular disease	Series 1	Manual counting	Self-paced treadmill walking and running.	No
Veerabhadrapa et al. ⁸⁰	71 (47); Range 18–55 years	Healthy adults	NR	Manual counting	Slow, moderate, brisk walking, and jogging.	No
Wallen et al. ⁴⁴	22 (11); 24.9 years (5.6)	Healthy adults	NR	Manual counting	Bruce graded treadmill test.	No
Sleep (3 studies)						
Lee et al. ⁸⁸	26 (NR); 43.6 years (14.1)	Individuals with subjective sleep discomfort recruited from hospital and clinic settings.	Series 8	Polysomnography	Single-night recording in a sleep laboratory.	No
Robbins et al. ⁸⁶	35 (15); Mean age NR	Healthy adults	Series 8	Polysomnography	Single-night recording in a sleep laboratory.	No
Apple ¹⁴⁶	166 (88); 47 years (16.5)	Healthy adults and those with sleep disorders	NR	Polysomnography, and home-based EEG recording with Prodigy Sleep System (Cerebra Medical Ltd.)	1 to 3 nights of recording in a sleep laboratory with PSG, or at home with EEG.	Yes
Sleep apnoea detection (1 study)						
Apple ²⁷	1499 (652); 46 years (14)	Healthy adults and those with sleep apnoea of different severity	NR	Home Sleep Apnea Tests (HSAT)	Participants wore Apple Watch for up to 30 nights and underwent a minimum of two nights of HSAT recordings with a type 3 HSAT device.	Yes
Heart rate variability (1 study)						
O'Grady et al. ⁴⁸	39 (17); 24.6 years (8.2)	Healthy adults	Series 9 and Ultra 2	Polar H10 + Kubios HRV software.	Simultaneous recordings each morning for 14 days.	Yes
Hypertension notification (1 study)						
Apple ¹⁴⁷	2229 (1268); Mean age NR	Adults with and without hypertension, and those with risk factors	NR	OMRON Evolv [®] Wireless Upper Arm Blood Pressure Monitor (Model BP7000, K162092).	Participants took blood pressure readings twice daily with the criterion and were instructed to wear Apple Watch for 12+ hours per day over 30 days.	Yes

Table 1 (continued) | Characteristics of included studies

Wheelchair push count (3 studies)						
VO ₂ max (1 study)						
Lambe et al. ⁸⁵	30 (15); 31.9 years (13.9)	Healthy adults	Series 5 to 9	Indirect calorimetry (Cosmed Quark CPET)	Treadmill running during maximal exercise test.	Yes
Six-Minute Walk Test distance (1 study)						
Apple ¹⁴⁸	449 (184); 78 years (7)	Adults aged 65 or older with various comorbidities, including diabetes, COPD, coronary artery disease, who used or didn't use an assistive walking device	Series 4	Supervised 6-min walk tests	Participants completed up to five supervised 6-min walk tests. They were then asked to wear their Apple Watch and carry their iPhone during normal day-to-day activities to generate a distance prediction.	Yes
Sound exposure (1 study)						
Fischer et al. ¹⁴⁹	1; Age NR	Healthy adults	Series 6	Class 1 sound level meter (XL2, NTI Audio with free-field measurement microphone (M2230-WP, NTI Audio))	Simultaneous noise measurement in 13 free-living environments.	No

SD standard deviation, NR not reported.
AFib atrial fibrillation, IQR interquartile range.

Detailed risk of bias assessment for each metric is provided as a supplementary file, with narrative synthesis in the Supplementary Information (pp. 2–3).

Heart rate

Thirty-eight studies (1855 participants; 66% male) validated heart rate measurements from all Apple Watch models through Series 9 and Ultra 2. Agreement with criterion measures was strongest at rest, whereas it was lower during exercise involving irregular movement patterns and among individuals with arrhythmia²⁹. Mean difference for resting heart rate ranged from -2.47 bpm to 3.61 bpm, and MAPE ranged from 1.69% to 7.2%^{30,31}. During exercise, 10/11 (91%) studies reported MAPE lower than 10%^{21,30,32–39}. MAPE tended to rise as intensity increased, although a decrease was noted in three studies^{32,34,38}.

Meta-analysis of heart rate, with resting and exercise conditions combined, included 22 studies ($n = 1247$)^{29–34,37,38,40–50}. The pooled mean bias (MB) was low, although limits of agreement (LoA) indicated measurement variability (-0.27 bpm [95% CI -0.72–0.17]; LoA -7.19 to 6.64; τ^2 0.53; Fig. 2). For resting heart rate, we found that Apple Watch measurements were higher than criterion measures (MB 0.21 bpm [95% CI -0.65–1.07]; LoA -8.14 to 8.56; τ^2 0.67; Fig. 3A). During exercise, Apple Watch underestimated heart rate (MB -0.63 bpm [95% CI -1.37–0.12]; LoA -6.86 to 5.60; τ^2 0.93; Fig. 3B).

Six studies (16%) were rated as 'low' risk of bias, 11 (29%) as 'some concerns', and 21 (55%) as 'high'. To examine the robustness of our findings, we conducted sensitivity analysis excluding studies at high risk of bias. The pooled mean bias and limits of agreement were comparable to our primary analysis (MB -0.50 bpm [95% CI -1.47–0.47]; LoA -7.54 to 6.53; 13 studies; Fig. S1).

To compare findings across Apple Watch models, we performed exploratory subgroup analysis according to the generation of optical heart rate sensor: first-generation (Apple Watch models up to Series 3), second-generation (Series 4–5 and all SE models), and third-generation (Series 6 onwards, including Ultra models). Compared to our primary analysis, we found narrower limits of agreement for the third-generation sensor (LoA -3.68 to 2.59; 8 studies; Fig. S2), but wider limits of agreement for the first- and second-generation sensors. Mean bias was comparable across all analyses. Further detail is provided in Supplementary Note 1.

Atrial fibrillation detection

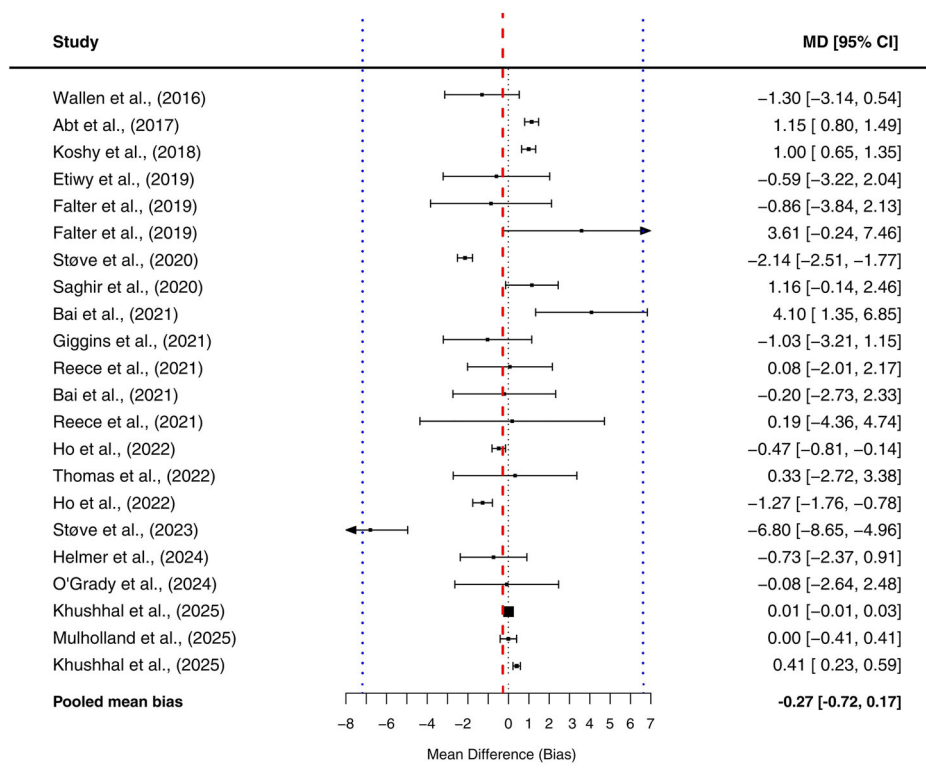
Seventeen studies validated atrial fibrillation detection ($n = 422,654$; 57% male): two evaluated PPG-based detection from tachograms (Irregular Rhythm Notification)^{25,50}, and the remainder assessed the ECG app. Sensitivity and specificity ranged widely between studies (19%–100% and 66%–100%, respectively). Six of the 15 studies that calculated sensitivity reported values higher than 80%^{51–56}, and six fell in the range of 65% to 90%^{50,57–61}. Sensitivity and specificity substantially improved when inconclusive ECG tracings were excluded^{51,53,56,59,60,62}. The rate of inconclusive tracings was between 15 and 25% in several studies^{52–55,60,63}. Thirteen studies were rated as 'high' risk of bias and four as 'some concerns'.

Eleven studies ($n = 3144$) were included in meta-analysis of atrial fibrillation detection, all of which validated the ECG app^{51–53,55–57,59,60,62–64}. Pooled sensitivity was 0.79 (95% CI 0.61–0.90), and pooled specificity was 0.91 (95% CI 0.81–0.96). The overall Zhou and Dendukuri I^2 indicated moderate heterogeneity (55%). The area under the curve suggested strong discriminative ability (0.93; Fig. 4). Exploratory subgroup analysis examining the influence of hardware and software version is presented in the Supplementary Information (p. 6).

ECG waveform morphology

Seven studies ($n = 535$, 68% male) compared the amplitude and duration of Apple Watch ECG recordings to 12-lead ECG^{46,49,65–69}. QT interval was the most frequently assessed segment (five studies)^{42,65,67–69}. Four studies reported that Apple Watch underestimated QT interval duration, although

Fig. 2 | Forest plot for heart rate under all conditions. The red dashed line represents the pooled mean bias; the blue dashed lines represent the pooled limits of agreement (-7.19 to 6.64).



limits of agreement were relatively wide^{42,65,67,68}. Many of these studies evaluated different segments of the ECG waveform, restricting comparison.

Blood oxygen saturation

Blood oxygen saturation (SpO₂) measurements were validated in Series 6 through Series 8, and six studies included patient cohorts^{42,45,70–73}. Seven studies reported overall mean difference <1% SpO₂, indicating good measurement accuracy, particularly in normoxic ranges^{42,70,72–76}. However, limits of agreement approximating $\pm 5\%$ SpO₂ were reported in multiple studies, indicating variability in measurements^{45,70,72,74–77}. Measurement error tended to increase as SpO₂ decreased. All five studies that assessed SpO₂ in both hypoxic and normoxic ranges found stronger agreement with criterion measures in normoxic ranges^{72,74–77}. Apple's white paper reported accuracy root mean square (A_{rms}) within the limits (<3.5%) defined by the US Food and Drug Administration (FDA) for medical pulse oximeters across the entire range of 70–100% SpO₂. Two additional studies also reported A_{rms} within these limits across the range of 80–100%^{75,76}. Contrastingly, two studies reported wide limits of agreement for hypoxic ranges, reflecting variability in accuracy^{72,77}.

Nine studies ($n = 969$) were included in meta-analysis of blood oxygen saturation. Pooled mean bias indicated that Apple Watch underestimated SpO₂, although limits of agreement demonstrated variability (MB -0.04% [95% CI -0.42–0.35], LoA -4.01 to 3.94; τ^2 0.13; Fig. 5). Our exploratory subgroup analysis found overestimation and wider limits of agreement for measurements obtained in hypoxic ranges (MB 0.43% [95% CI -3.85–4.71]; LoA -8.35 to 9.21; Supplementary Information p. 7).

Energy expenditure

Margins of error for energy expenditure estimates were often large, both during exercise and at rest (8 studies; $n = 270$; 63% male). There was considerable variation between and within individual studies. Participants were predominantly young physically active adults, and five of the eight studies assessed Apple Watch Series 2 or older. All six studies that calculated MAPE reported values of 20% or higher in at least one test condition^{31,32,36,39,78,79}. Overall, MAPE ranged from 9.71%

(running) to 151.66% (walking). No distinct trend in measurement error by exercise intensity could be observed.

Step count and wheelchair push count

Three studies validated step count from Apple Watch First Generation and Series 1. In the largest study ($n = 71$), a small underestimation and strong correlation was found, however, moderate correlation and wide limits of agreement were reported in each of the other studies⁸⁰. There was no distinct trend in accuracy based on walking or running speed^{80,81}. Notably, no study included sedentary periods or seated activities that involved arm movements in their validation).

Three studies evaluated wheelchair push count. Apple Watch overestimated overall wheelchair push count in two studies^{82,83}, and underestimated in the other⁸⁴. However, margins of error varied substantially, even within studies. MAPE ranged from 1% to 21% for Series 1^{83,84}, and was 9.2% for Series 4⁸².

VO₂ max estimation

One study ($n = 30$) compared VO₂ max estimates to indirect calorimetry and found that Apple Watch underestimated VO₂ max, noting a clinically significant mean difference (-6.07 mL/kg/min) and wide limits of agreement⁸⁵.

Sleep stage classification and sleep apnoea detection

Three studies validated sleep stage classification ($n = 221$)^{86–88}. Overall, they found good differentiation between sleep and wake states, but moderate-to-poor differentiation between physiologically similar sleep stages. Two studies reported sensitivity for binary sleep-wake classification $\geq 97\%$, however, they also reported low accuracy for classification of deep sleep, with a tendency to misclassify it as light sleep^{86,87}. Robbins and colleagues ($n = 29$, Series 8) found that Apple Watch significantly underestimated deep sleep, and overestimated light sleep⁸⁶. For sleep apnoea detection, Apple's clinical validation study found higher specificity (98.5% [95% CI 98.0–99.0]) than sensitivity (66.3% [95% CI 62.2–70.3]). Fig. 6 provides a graphical overview of this review's results.

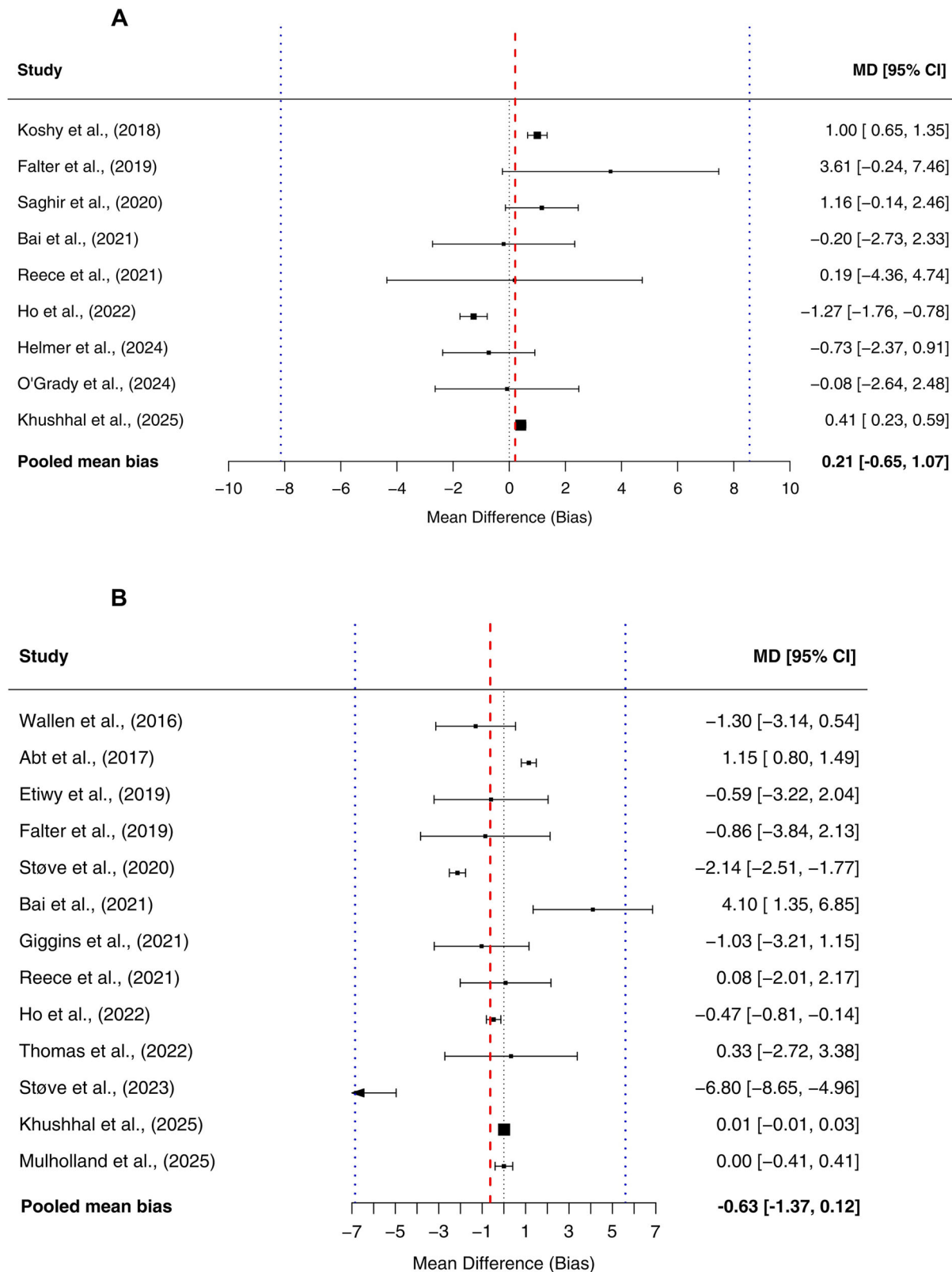


Fig. 3 | Forest plots of heart rate at rest and during exercise. **A** Forest plot for heart rate at rest. The pooled mean bias (0.21 bpm) and limits of agreement (-8.14 to 8.56) are represented by the dashed red and blue lines, respectively. **B** Forest plot for heart rate during exercise (mean bias -0.63; limits of agreement -6.86 to 5.60).

Discussion

This systematic review and meta-analysis evaluated the accuracy of 14 health metrics from Apple Watch to inform its use in personal health monitoring and clinical settings. We found that accuracy varied by metric, measurement conditions, and physiological

characteristics, highlighting the need to interpret accuracy in the context of each metric's intended use.

The pooled mean bias for heart rate was low (-0.27 bpm [95% CI -0.72–0.17]), although limits of agreement were moderately wide (-7.19 to 6.64 bpm). The pooled limits of agreement demonstrated measurement

Fig. 4 | Summary Receiver Operating Characteristic Curve for atrial fibrillation detection.

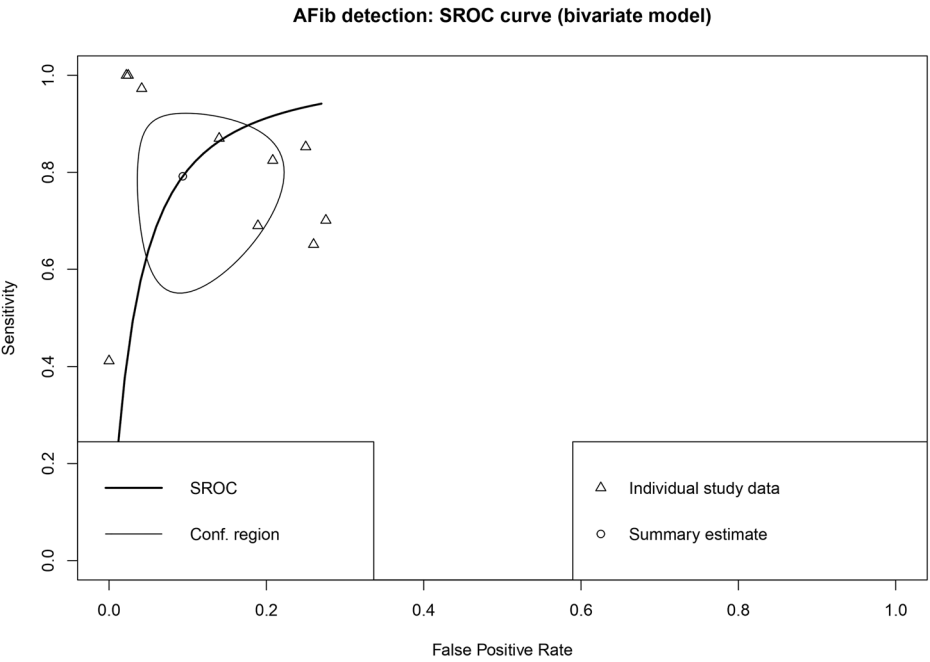
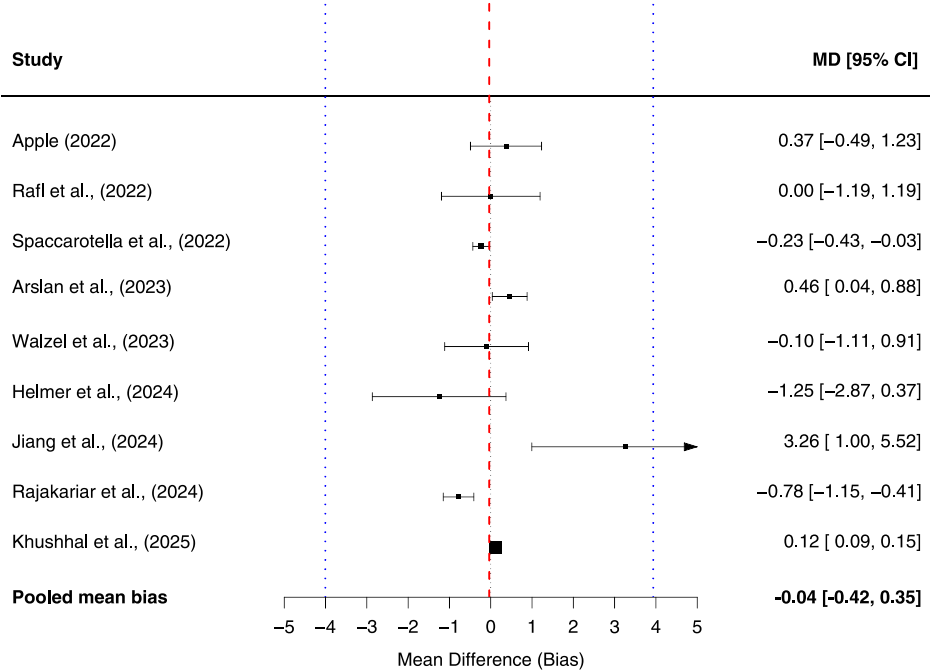


Fig. 5 | Forest plot of blood oxygen saturation measurement accuracy. The pooled mean bias (-0.04% SpO₂) and limits of agreement (-4.01 to 3.94) are represented by the dashed red and blue lines, respectively.



variability of $\sim\pm 7$ bpm and reflected agreement across a broad population by incorporating both within- and between-study variability, as described by Tipton & Shuster. In line with Bland and Altman’s recommendations, the limits of agreement are the key measure for determining whether Apple Watch is a suitable alternative to current measurement methods. We observed sufficient accuracy to quantify exercise intensity among healthy adults, although moderate misestimation may occur in some cases, particularly among individuals with cardiac disease. Our subgroup analyses showed substantially lower variability for measurements obtained with the third-generation optical sensor (LoA -3.68 to 2.59) compared to older generations. This indicated that accuracy was both population- and condition-dependent.

For blood oxygen saturation, we also found low mean bias (-0.04% [95% CI -0.42–0.35]), but the pooled limits of agreement (-4.01 to 3.94) suggested that Apple Watch may, in certain instances, misclassify individuals in hypoxic ranges as being in normoxic ranges. Across individual studies and in our subgroup analysis, we identified greater variability and lower agreement among patients in hypoxaemia. However, two studies found that, in healthy adults, Apple Watch met the standards set by the FDA and International Organization for Standardization (ISO) for medical grade pulse oximetry when hypoxaemia was induced. These findings indicate that Apple Watch may serve as a useful adjunct to traditional pulse oximetry, although its accuracy is limited in hypoxic ranges.

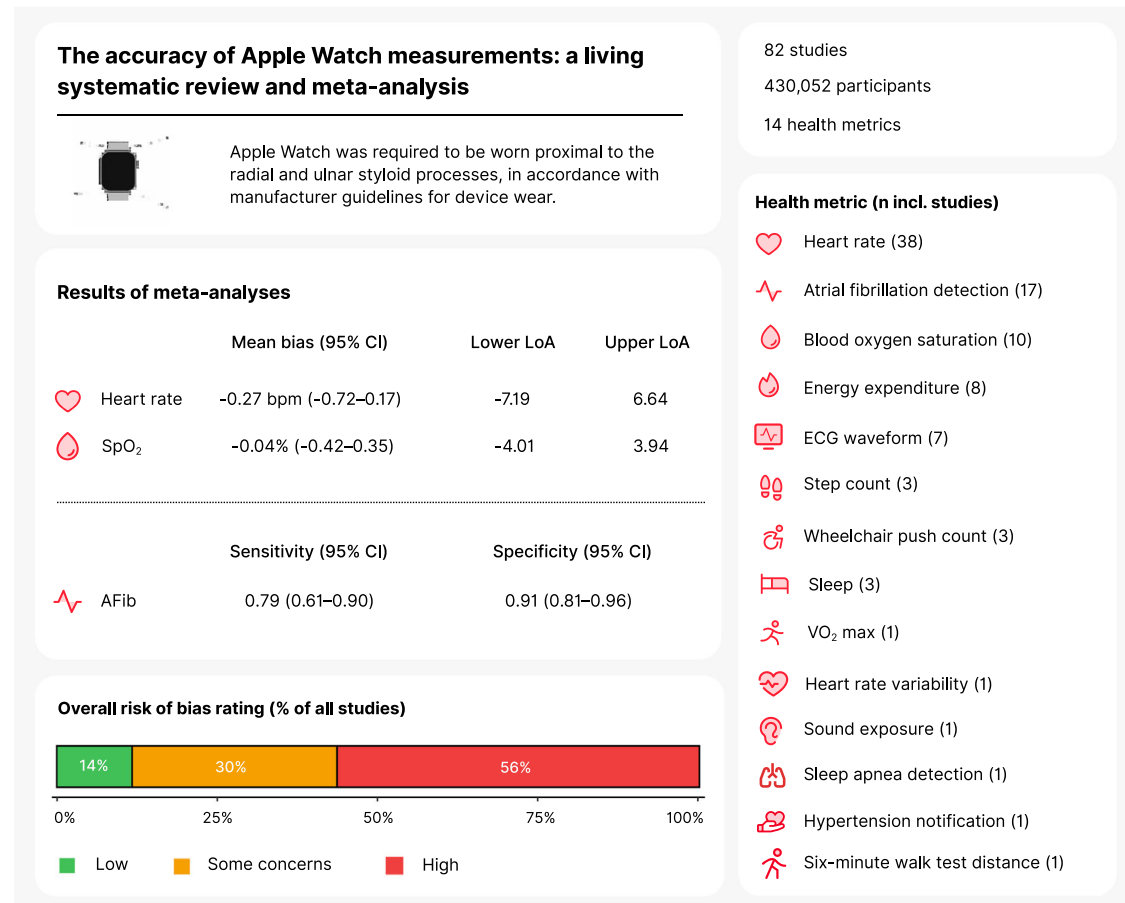


Fig. 6 | Graphical abstract. Demonstrating included metrics, inclusion requirement for device wear, risk of bias ratings, and meta-analysis results. bpm beats per minute, LoA limits of agreement. Icons adapted from Phosphor Icons, used under the MIT License.

For atrial fibrillation detection, Apple Watch was more specific than sensitive (pooled sensitivity 0.79 [95% CI 0.61–0.90]). The pooled specificity (0.91 [95% CI 0.81–0.96]) indicated that notification of atrial fibrillation likely reflects true presence, suggesting notification warrants further clinical investigation. Both sensitivity and specificity ranged widely between studies, however, and in many, more than a quarter of measurements were inconclusive, representing a notable rate of unsuccessful assessment.

The error of energy expenditure estimates was often large and varied considerably, both within and between studies, and the mean difference for VO₂ max (-6.07 mL/kg/min) was clinically significant, as a 3.5 mL/kg/min increase has been associated with a risk ratio of 0.89 for all-cause mortality⁸⁹. We observed moderate accuracy for sleep overall, with good classification between sleep and wake states — sufficient for personal health monitoring — but differentiation between physiologically similar sleep stages was poor. There was also moderate accuracy for step count, wheelchair push count and hypertension notification, although there were fewer than four studies included for each metric. A number of metrics are yet to be validated, including respiratory rate, wrist temperature and measures of sedentary behaviour.

There are important distinctions between our findings and previous systematic reviews and meta-analyses, although we report similar results for certain metrics^{22–24,90–96}. A prior meta-analysis, which pooled multiple effect estimates from single studies — a method that is not recommended⁹⁷ — found a similar mean bias but wider limits of agreement for heart rate (-0.12 bpm; LoA -11.06 to 10.81)²². Notably, the authors included several studies that we deemed ineligible for our review, primarily due to the validity of criterion methods and lack of adherence to manufacturer guidelines for device wear. Elsewhere, low

and moderate agreement have been identified for energy expenditure and step count, respectively^{22–24}. Many of these previous systematic reviews, however, included fewer than five studies and exclusively assessed old Apple Watch software and hardware^{23,24}. Only two prior meta-analyses have evaluated atrial fibrillation detection. The first pooled just three studies using a fixed-effects model, which does not appropriately account for heterogeneity⁹³, while the second meta-analysis pooled results from multiple manufacturers' devices⁹².

We found that Apple Watch's measurement accuracy broadly aligns with that of other wearable devices. Across manufacturers, error margins for energy expenditure estimates are often large⁹⁸, whereas heart rate measurements typically exhibit stronger agreement with criterion measures²⁶. For heart rate and blood oxygen saturation, Apple Watch showed stronger agreement with criterion measures than Garmin, Fitbit, and Withings devices^{23,24,71,99}. For sleep, however, agreement with polysomnography was lower for Apple Watch than for Whoop, Fitbit, and Garmin^{88,100}.

Three factors particularly impact measurement accuracy. Firstly, the metric's measurement method. Metrics such as step count, VO₂ max, and energy expenditure require inputs from multiple sensors, combined through sensor fusion²⁷. When they are combined, error from individual inputs may compound^{101,102}. In contrast, metrics like heart rate and SpO₂ are obtained directly from photoplethysmography (PPG), requiring less derivation. Secondly, factors such as movement, moisture, and skin contact impact motion sensor measurements and the clarity of PPG waveforms^{27,103,104}. This is one source of inaccurate heart rate measurements during high-intensity exercise with irregular movement patterns²⁹. Thirdly, physiological factors, including blood perfusion and individual variation in heart rate response to exercise affect measurements¹⁸. Low blood perfusion,

due to low body temperature or physiological traits, can lead to inaccuracy, especially given the PPG sensor's reliance on pulsatile arterial blood, which accounts for a minority of blood in the tissue at the wrist²⁷. Algorithms that are ill-suited to an individual's physiology may also lead to inaccuracy. Given the sensitivity of PPG waveforms and sensor measurements to these factors, the machine learning algorithms that interpret them are increasingly important, and recent literature has shown improved accuracy due to algorithmic developments alone¹⁰⁵.

To determine whether accuracy is adequate, the measurement's intended use must be considered. For clinical use, thresholds corresponding to clinically important change may guide interpretation. For instance, a 10 bpm increase in resting heart rate has been associated with a 9% increase in all-cause mortality risk¹⁶, whereas a 3.5 mL/kg/min increase in VO_2 max and a 1000-step increase in daily step count have both been associated with decreased all-cause mortality risk^{89,106}. Accuracy that permits detection of clinically meaningful change — within thresholds identified by large epidemiological studies and meta-analyses, or those stipulated by regulatory bodies such as the FDA, ISO, and European Union^{107–109} — may be deemed adequate. For personal health and fitness monitoring, however, wider margins of error may suffice to provide high-level trends over time in physiological and behavioural health metrics. In population-level research trials, where scale may attenuate individual error, such measurements could provide researchers with insight into associations and risk stratification across groups. The required accuracy, therefore, should be guided by the measurement's use and by validation among the intended measurement population.

We recognise that our results are contingent on the characteristics of our included studies, particularly given the variability in accuracy across participant cohorts and measurement conditions. A greater proportion of trials involving cardiac populations or exercise involving erratic movement patterns, for instance, may have produced different results. Methodological rigour was also inconsistent: adherence to validation guidelines, such as INTERLIVE's expert statements, was low^{110–113}, statistical procedures were sometimes inadequately described, and inconclusive measurements were excluded from certain analyses. In addition, few studies conducted free-living validation, which best reflects typical use, likely due to challenges obtaining criterion measures.

Consequently, our study has several limitations. First, statistical and methodological heterogeneity prevented meta-analysis of energy expenditure, and restricted subgroup analyses. We were unable to conduct subgroup analysis by body mass index or skin tone as it was infrequently reported. Additionally, we could not precisely differentiate between the impact of hardware and software on accuracy due to the proprietary nature of updates to the foreground heart rate and SpO_2 algorithms, as well as the limited number of studies evaluating each Apple Watch model. Second, the generalisability of our findings was restricted due to the bias towards physically active individuals and males among participants. The variation in sex balance between metrics, coupled with limited validation among older adults and those with comorbidities, accentuates this restriction. Third, many studies were at high risk of bias. While we conducted sensitivity analyses excluding these studies for heart rate, this was not feasible for blood oxygen saturation and atrial fibrillation; fewer than five studies were rated as 'low' or 'some concerns' for these metrics, and the marked imbalance between groups would have limited the validity and interpretability of any formal analysis. Fourth, few studies were included for metrics such as step count and sleep. This was due to our stringent approach to criterion method validity and adherence to manufacturer guidelines for device wear. Fifth, many studies assessed Apple Watch models that have since been discontinued. Nevertheless, several studies validated measurements from the most recent optical heart rate sensor and algorithms, as they are not updated with each new Apple Watch model.

The main strength of this study is its breadth and meta-analyses. It is the first to synthesise all health metrics from Apple Watch that have currently been validated, and it provides the most comprehensive meta-analyses to date of heart rate, atrial fibrillation detection, and blood oxygen saturation. We gave ample consideration to the validity of criterion methods

and ensured that Apple Watch was validated in the manner it was designed to be worn. We did not consider research-grade wearables as valid criterion methods for step count or energy expenditure due to the conflicting evidence on their validity^{98,114,115}. A rigorous search and screening process was implemented, comprising nine databases and four reviewers, and to reduce publication bias, grey literature was included. This study is designed as a living systematic review and meta-analysis to ensure that the evidence synthesis does not become outdated quickly as Apple Watch evolves. An updated search will be conducted yearly to integrate new studies and new metrics, and data will be published in an open-access format.

The clinical applications of wearable devices are budding. There is growing recognition that wearable devices may improve preventative care and management of chronic disease^{2,102}. Major organisations, including the American Heart Association and the British Heart Foundation, are conducting large research trials to inform the integration of wearable data in cardiovascular care^{2,116–118}. Moreover, the development of digital biomarkers, together with emerging metrics such as hypertension notification, aim to translate wearable measurements into clinically actionable data that support disease management and assessment. Clear interpretation of these data may provide agency to patients, allowing them to better manage their condition in partnership with their healthcare professional, ultimately reducing health-care cost and burden^{102,119–122}.

Future research should examine the longitudinal relationships of Apple Watch metrics with markers of health and disease, as well as validating measurements taken at single time-points. Clearer understanding of measurement precision and reliability will enable more accurate interpretation of trends in health metrics over time. Validation studies that include older adults, patient populations, and metrics related to vital signs — such as respiratory rate and wrist temperature — are needed. As software and hardware advance, and new metrics are developed, continued validation across diverse cohorts and conditions is required to inform the capabilities and limitations of Apple Watch.

This systematic review and meta-analysis demonstrated the variation in measurement accuracy between Apple Watch health metrics, as well as the influence of measurement condition and individual physiology. We identified good agreement for heart rate overall, whereas error for energy expenditure estimates was often inconsistent and large. Wide limits of agreement for SpO_2 indicated measurement variability, and we found moderate accuracy for sleep and step count. As a ubiquitous consumer device, Apple Watch provides the general population with assessment of activity, physiology, and cardiovascular function that may otherwise be inaccessible. Despite inaccuracies, the continuous nature of these measurements may offer unique health insights, and further research exploring their use in public health is warranted.

Methods

This systematic review and meta-analysis was conducted and reported as per PRISMA guidelines¹²³. The protocol was prospectively registered in PROSPERO (CRD42023481841; www.crd.york.ac.uk/PROSPERO/view/CRD42023481841).

Search strategy and selection criteria

We searched PubMed, SPORTDiscus, Embase, IEEE Xplore, Web of Science, Scopus, CINAHL and the Cochrane Library from inception to September 24, 2025. Keywords, Medical Subject Headings (MeSH), and synonyms related to Apple Watch and its measurement accuracy were included. To identify additional studies and grey literature, a hand search was undertaken across Google Scholar, the Apple Health website, and the US Food and Drug Administration 510(k) database. The university's Research Engagement Librarian was involved throughout the development of the search strategy, which was peer-reviewed prior to implementation. Details of the tailored search strategy for each database are reported in Supplementary Note 2.

We included primary research studies which compared any health metric from Apple Watch to a validated criterion measure. Description of

valid criterion measures are available in the Supplementary Information (pp. 11–12). Studies investigating metrics not intended to be measured by Apple Watch, or in populations in which they were not intended for use, were excluded; for example, recording ECG with Apple Watch placed at the ankle, or blood oxygen saturation assessment in neonates. Measurements were required to be taken in accordance with manufacturer guidelines. Studies in which multiple devices were worn on one wrist were excluded due to potential measurement interference caused by improper device placement, photoplethysmographic light impedance from adjacent devices, and motion sensor disruption, among other factors. Grey literature, including conference abstracts and unpublished white papers, was also included. There were no restrictions placed on demographic or language.

Three authors (RL, B.O.G., M.B.) independently screened titles, abstracts, and full texts, with two authors per citation. Disagreements were resolved by consensus. The study selection process was carried out using Covidence (Veritas Health Innovation Ltd). This study was designed as a living systematic review. Searches will be updated every 12 months, or earlier if major Apple Watch hardware or software updates occur. Newly identified studies will be screened and incorporated using the same methodology. Updates will be disseminated via the Open Science Framework (osf.io/v5d3k).

Outcomes

The primary outcome was the agreement between measurements from Apple Watch and the criterion method for each health metric. This included pooled mean bias, Bland-Altman limits of agreement, sensitivity, and specificity for metrics that were meta-analysed. We extracted measures of agreement across all populations and conditions, including varied exercise intensities and clinical cohorts (e.g., cardiovascular disease). Measures of effect included mean difference, sensitivity and specificity, mean average percentage error (MAPE), Bland-Altman limits of agreement, and correlation coefficients.

Data extraction

Two reviewers (RL, M.B.) independently extracted data in duplicate using a pilot-tested extraction form in Microsoft Excel. Extracted data were then compared and merged following consensus. This included data about participant demographic, criterion method, validation protocol, and statistical analysis. In the case of missing or unclear information, authors were contacted via email, and one follow-up was sent to those who did not respond. Where required, we back-calculated statistics necessary for meta-analysis, if sufficient data were available¹²⁴.

Risk of bias assessment

An adapted version of the COSMIN checklist (COnsensus-based Standards for the selection of health Measurement INstruments) was used to assess risk of bias. COSMIN defines standards for evaluating the methodological quality of studies validating health measurement instruments and is implemented by the expert-led 'Towards Intelligent Health and Well-Being Network of Physical Activity Assessment' (INTERLIVE) consortium^{110,125}. The modified tool includes four domains: participants, index measure, reference standard, and statistical analysis. Each domain includes multiple items with three possible answers ('yes', 'unclear', or 'no'), and ratings were assigned in accordance with the checklist's recommendations. Studies with at least one 'no', or more than two 'unclear' ratings were categorised as 'high' risk, while those with one 'unclear' item were designated as 'some concerns'. Studies with 'yes' in all domains were classified as 'low risk'. Where studies validated more than one metric, risk of bias was assessed individually for each. Three authors (RL, B.O.G., M.B.) independently assessed risk of bias and disagreements were resolved by consensus.

Statistical analysis

Meta-analysis of heart rate and blood oxygen saturation was conducted in accordance with the framework developed by Tipton & Shuster¹²⁶. A random-effects model with inverse variance weighting was used account for

heterogeneity between trials¹²⁷. Pooled Bland-Altman limits of agreement and mean bias were calculated. Subgroup meta-analyses were conducted for heart rate measured at rest and during exercise. To prevent unit-of-analysis errors, only one estimate per study per condition was included in meta-analyses, in line with the approach described by Borenstein and colleagues⁹⁷. Where studies reported multiple mean difference values, they were pooled prior to meta-analysis, accounting for variance. If the standard deviation of the differences was not reported, it was back-calculated by rearranging the formula used to compute 95% limits of agreement¹²⁸. Details of the formulae for back-calculation and the methods for pooling mean differences are provided in the Supplementary Information (p. 15).

Pooled sensitivity and specificity for atrial fibrillation detection was calculated using bivariate meta-analysis with *Reitsma* (*mada* package)¹²⁹. Diagnostic accuracy contingency tables were back-calculated when not reported, in accordance with previously described methods (appendix p. 14)¹²⁴. We evaluated statistical heterogeneity by estimating the degree of between-study variability using the Tau² statistic^{130,131}. Analyses were conducted in R version 4.5.1 (The R Foundation for Statistical Computing, Vienna) with RStudio (version 2025.09.0 + 387) and in Python 3.13

Data availability

Synthesised results data, risk of bias assessment, and study protocol are available via the Open Science Framework, at osf.io/v5d3k. Raw datasets generated as part of this review are available from the corresponding author upon request.

Code availability

The code used in this study is publicly available via GitHub, at github.com/rorylambe/applewatch-systematicreview. This repository includes R code used to conduct meta-analyses of heart rate, atrial fibrillation, and blood oxygen saturation.

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References

- Smuck, M., Odonkor, C. A., Wilt, J. K., Schmidt, N. & Swiernik, M. A. The emerging clinical role of wearables: factors for successful implementation in healthcare. *NPJ Digit. Med.* **4**, 45 (2021).
- Truslow, J. et al. Understanding activity and physiology at scale: the Apple heart & movement study. *npj Digit. Med.* **7**, 242 (2024).
- Shapiro, I., Stein, J., MacRae, C. & O'Reilly, M. Pulse oximetry values from 33,080 participants in the Apple Heart & Movement Study. *NPJ Digit. Med.* **6**, 134 (2023).
- Bent, B. et al. Engineering digital biomarkers of interstitial glucose from noninvasive smartwatches. *NPJ Digit. Med.* **4**, 89 (2021).
- Ballinger, B. et al. DeepHeart: Semi-supervised sequence learning for cardiovascular risk prediction. In *Proc. AAAI Conference on Artificial Intelligence*, 32 (AAAI, 2018).
- Pammi, M. et al. Digital twins, synthetic patient data, and in-silico trials: can they empower paediatric clinical trials? *Lancet Digit. Health* **7**, 100851 (2025).
- Gibson, C. M. et al. Does early detection of atrial fibrillation reduce the risk of thromboembolic events? rationale and design of the Heartline study. *Am. Heart J.* **259**, 30–41 (2023).
- Mandsager, K. et al. Association of cardiorespiratory fitness with long-term mortality among adults undergoing exercise treadmill testing. *JAMA Netw. Open* **1**, e183605 (2018).
- Jayed, A., Gohari, A. & Shab-Bidar, S. Daily step count and all-cause mortality: a dose-response meta-analysis of prospective cohort studies. *Sports Med.* **52**, 89–99 (2022).
- Kokkinos, P. et al. Cardiorespiratory fitness and mortality risk across the spectra of age, race, and sex. *J. Am. Coll. Cardiol.* **80**, 598–609 (2022).

11. Clausen, J. S., Marott, J. L., Holtermann, A., Gyntelberg, F. & Jensen, M. T. Midlife cardiorespiratory fitness and the long-term risk of mortality: 46 years of follow-up. *J. Am. Coll. Cardiol.* **72**, 987–995 (2018).
12. Tsuji, H. et al. Reduced heart rate variability and mortality risk in an elderly cohort. The Framingham Heart Study. *Circulation* **90**, 878–883 (1994).
13. Jarczok, M. N. et al. Heart rate variability in the prediction of mortality: a systematic review and meta-analysis of healthy and patient populations. *Neurosci. Biobehav. Rev.* **143**, 104907 (2022).
14. Gallicchio, L. & Kalesan, B. Sleep duration and mortality: a systematic review and meta-analysis. *J. sleep. Res.* **18**, 148–158 (2009).
15. Kwok, C. S. et al. Self-reported sleep duration and quality and cardiovascular disease and mortality: a dose-response meta-analysis. *J. Am. Heart Assoc.* **7**, e008552 (2018).
16. Zhang, D., Shen, X. & Qi, X. Resting heart rate and all-cause and cardiovascular mortality in the general population: a meta-analysis. *Cmaj* **188**, E53–E63 (2016).
17. Dohrn, M., Sjöström, M., Kwak, L., Oja, P. & Hagströmer, M. Accelerometer-measured sedentary time and physical activity—a 15 year follow-up of mortality in a Swedish population-based cohort. *J. Sci. Med. Sport* **21**, 702–707 (2018).
18. Bent, B., Goldstein, B. A., Kibbe, W. A. & Dunn, J. P. Investigating sources of inaccuracy in wearable optical heart rate sensors. *NPJ digital Med.* **3**, 18 (2020).
19. Alugubelli, N., Abuissa, H. & Roka, A. Wearable devices for remote monitoring of heart rate and heart rate variability-what we know and what is coming. *Sensors (Basel)* **22**, 8903 (2022).
20. Hajj-Boutros, G., Landry-Duval, M.-A., Comtois, A. S., Gouspillou, G. & Karelis, A. D. Wrist-worn devices for the measurement of heart rate and energy expenditure: a validation study for the Apple Watch 6, Polar Vantage V and Fitbit Sense. *Eur. J. Sport Sci.* **23**, 165–177 (2023).
21. Kim, C., Song, J. H. & Kim, S. H. Validation of wearable digital devices for heart rate measurement during exercise test in patients with coronary artery disease. *Ann. Rehabil. Med.* **47**, 261–271 (2023).
22. Choe, J. P. & Kang, M. Apple Watch accuracy in monitoring health metrics: a systematic review and meta-analysis. *Physiol. Meas.* <https://doi.org/10.1088/1361-6579/adca82> (2025).
23. Fuller, D. et al. Reliability and validity of commercially available wearable devices for measuring steps, energy expenditure, and heart rate: systematic review. *JMIR Mhealth Uhealth* **8**, e18694 (2020).
24. Germini, F. et al. Accuracy and acceptability of wrist-wearable activity-tracking devices: systematic review of the literature. *J. Med. Int. Res* **24**, e30791 (2022).
25. Perez, M. V. et al. Large-scale assessment of a smartwatch to identify atrial fibrillation. *N. Engl. J. Med.* **381**, 1909–1917 (2019).
26. Doherty, C., Baldwin, M., Keogh, A., Caulfield, B. & Argent, R. Keeping pace with wearables: a living umbrella review of systematic reviews evaluating the accuracy of consumer wearable technologies in health measurement. *Sports Med.* **54**, 1–20 (2024).
27. Apple. *Using Apple Watch To Measure Heart Rate, Calorimetry, And Activity.* <https://www.apple.com> (2024).
28. Goldsack, J. C. et al. Verification, analytical validation, and clinical validation (V3): the foundation of determining fit-for purpose for Biometric Monitoring Technologies (BioMeTs). *npj Digit. Med.* **3**, 55 (2020).
29. Stove, M. P. & Hansen, E. C. K. Accuracy of the Apple Watch Series 6 and the Whoop Band 3.0 for assessing heart rate during resistance exercises. *J. Sports Sci.* **40**, 2639–2644 (2022).
30. Reece, J. D., Bunn, J. A., Choi, M. & Navalta, J. W. Assessing heart rate using consumer technology association standards. *Technologies* **9**, 46 (2021).
31. Bai, Y. et al. Comprehensive comparison of Apple Watch and Fitbit monitors in a free-living setting. *PLoS ONE* **16**, e0251975 (2021).
32. Falter, M., Budts, W., Goetschalckx, K., Cornelissen, V. & Buys, R. Accuracy of Apple Watch measurements for heart rate and energy expenditure in patients with cardiovascular disease: cross-sectional study. *JMIR mHealth uHealth* **7**, e11889 (2019).
33. Giggins, O. M. et al. In *43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society EMBC*, 6970–6973 (IEEE, 2021).
34. Ho, W.-T., Yang, Y.-J. & Li, T.-C. Accuracy of wrist-worn wearable devices for determining exercise intensity. *Digi. Health* **8**, 20552076221124393 (2022).
35. Nelson, B. W. & Allen, N. B. Accuracy of consumer wearable heart rate measurement during an ecologically valid 24 hour period: intraindividual validation study. *JMIR mHealth uHealth* **7**, e10828 (2019).
36. Nuss, K. J. et al. Accuracy of heart rate and energy expenditure estimations of wrist-worn and arm-worn Apple watches. *J. Meas. Phys. Behav.* **2**, 166–175 (2019).
37. Støve, M. P. et al. Measurement latency significantly contributes to reduced heart rate measurement accuracy in wearable devices. *J. Med. Eng. Technol.* **44**, 125–132 (2020).
38. Thomas, J., Doyle, P. & Doyle, J. A. Validity of optical heart rate measurement in commercially available wearable fitness tracking devices. *bioRxiv* **9**, e77911 (2022).
39. Uphill, A. et al. Validity of Apple Watch, Garmin Forerunner® 935 and GENEActiv for estimating energy expenditure during close quarter battle training in Special Forces soldiers. *Eur. J. Sport Sci.* **24**, 614–622 (2024).
40. Abt, G., Bray, J. & Benson, A. C. The validity and inter-device variability of the Apple Watch™ for measuring maximal heart rate. *J. Sports Sci.* **36**, 1447–1452 (2018).
41. Etiwy, M. et al. Accuracy of wearable heart rate monitors in cardiac rehabilitation. *Cardiovasc. Diagnosis Ther.* **9**, 262 (2019).
42. Khushhal, A. A., Mohamed, A. A. & Elsayed, M. E. Accuracy of apple watch to measure cardiovascular indices in patients with cardiac diseases: observational study. *Glob. Heart* **20**, 74 (2025).
43. Mulholland, A. M., MacDonald, H. V., Aguiar, E. J. & Wingo, J. E. Influence of skin pigmentation on the accuracy and data quality of photoplethysmographic heart rate measurement during exercise. *Eur. J. Appl. Physiol.* <https://doi.org/10.1007/s00421-025-05977-x> (2025).
44. Wallen, M. P., Gomersall, S. R., Keating, S. E., Wisløff, U. & Coombes, J. S. Accuracy of heart rate watches: implications for weight management. *PLoS ONE* **11**, e0154420 (2016).
45. Helmer, P. et al. Reliability of continuous vital sign monitoring in post-operative patients employing consumer-grade fitness trackers: A randomised pilot trial. *Digit. Health* **10**, 20552076241254026 (2024).
46. Khushhal, A. A., Mohamed, A. A., Alsegame, M. M. & Alsaedi, A. M. Accuracy of Apple Watch in measuring 30-second resting electrocardiography in patients with cardiac diseases and comorbidity: an observational cross-sectional study. *J. Multidiscip. Healthcare* **18**, 493–504 (2025).
47. Koshy, A. N. et al. Smart watches for heart rate assessment in atrial arrhythmias. *Int. J. Cardiol.* **266**, 124–127 (2018).
48. O'Grady, B., Lambe, R., Baldwin, M., Acheson, T. & Doherty, C. The validity of apple watch series 9 and ultra 2 for serial measurements of heart rate variability and resting heart rate. *Sensors* <https://doi.org/10.3390/s24196220> (2024).
49. Saghir, N. et al. A comparison of manual electrocardiographic interval and waveform analysis in lead 1 of 12-lead ECG and apple watch ECG: A validation study. *Cardiovasc. Digit Health J.* **1**, 30–36 (2020).

50. Wasserlauf, J. et al. Accuracy of the Apple watch for detection of AF: A multicenter experience. *J. Cardiovasc. Electrophysiol.* **34**, 1103–1107 (2023).
51. Abu-Alrub, S. et al. Smartwatch electrocardiograms for automated and manual diagnosis of atrial fibrillation: a comparative analysis of three models. *Front. Cardiovasc. Med.* **9**, 836375 (2022).
52. Mannhart, D. et al. Clinical validation of an artificial intelligence algorithm offering cross-platform detection of atrial fibrillation using smart device electrocardiograms. *Arch. Cardiovasc. Dis.* **116**, 249–257 (2023).
53. Mannhart, D. et al. Clinical validation of 5 direct-to-consumer wearable smart devices to detect atrial fibrillation: BASEL wearable study. *Clin. Electrophysiol.* **9**, 232–242 (2023).
54. Scholten, J. et al. A comparison of over-the-counter available smartwatches and devices for electrocardiogram based detection of atrial fibrillation. *Eur. Heart J. Digit. Health* **2**, ztab104. 3047 (2021).
55. Velraeds, A. et al. Improving automatic smartwatch electrocardiogram diagnosis of atrial fibrillation by identifying regularity within irregularity. *Sensors* **23**, 9283 (2023).
56. Apple. *Using Apple Watch for Arrhythmia Detection*. https://www.apple.com/healthcare/docs/site/Apple_Watch_Arrhythmia_Detection.pdf (2020).
57. Briosa, E. G. A. et al. Diagnostic performance of single-lead electrocardiograms from a smartwatch and a smarting for cardiac arrhythmia detection. *Heart Rhythm* **6**, 808–817 (2025).
58. Müller, M. et al. Validity of a smartwatch for detecting atrial fibrillation in patients after heart valve surgery: a prospective observational study. *Scand. Cardiovasc. J.* **58**, 2353069 (2024).
59. Peppinkhuizen, S. et al. Accuracy and clinical relevance of the single-lead Apple Watch electrocardiogram to identify atrial fibrillation. *Cardiovasc. Digit. Health J.* **3**, S17–S22 (2022).
60. Racine, H.-P. et al. Role of coexisting ECG anomalies in the accuracy of smartwatch ECG detection of atrial fibrillation. *Can. J. Cardiol.* **38**, 1709–1712 (2022).
61. Saghir, N. S. et al. Correlation of atrial fibrillation detection using our ring with photoplethysmography in comparison to the apple watch electrocardiography algorithm (DH-576-04). *Heart Rhythm* **19**, S61–S62 (2022).
62. Ford, C., Xie, C. X., Low, A., Roberts, L. & Teh, A. W. Smart wars-comparison of the apple watch series 4 and kardiaband smart watch technology for the diagnosis of atrial fibrillation. *J. Am. Coll. Cardiol.* **77**, 3226–3226 (2021).
63. Seshadri, D. R. et al. Accuracy of Apple Watch for detection of atrial fibrillation. *Circulation* **141**, 702–703 (2020).
64. Wouters, F. et al. Comparative evaluation of consumer wearable devices for atrial fibrillation detection: validation study. *JMIR Formative Res.* **9**, e65139 (2025).
65. Behzadi, A. et al. Feasibility and reliability of smartwatch to obtain 3-lead electrocardiogram recordings. *Sensors* **20**, 5074 (2020).
66. Buelga Suárez, M. et al. Smartwatch ECG tracing and ischemic heart disease: ACS watch study. *Cardiology* **148**, 78–82 (2023).
67. Harmon, D. et al. Performance and accuracy of a smart watch single-lead ecg: a pilot study (Po-626-01). *Heart Rhythm* **19**, S150 (2022).
68. Klier, K., Koch, L., Graf, L., Schinköthe, T. & Schmidt, A. Diagnostic accuracy of single-lead electrocardiograms using the Kardia Mobile App and the Apple Watch 4: validation study. *JMIR Cardiol* **7**, e50701 (2023).
69. Strik, M. et al. Validating QT-interval measurement using the Apple Watch ECG to enable remote monitoring during the COVID-19 pandemic. *Circulation* **142**, 416–418 (2020).
70. Arslan, B. et al. Accuracy of the apple watch in measuring oxygen saturation: comparison with pulse oximetry and ABG. *Ir. J. Med. Sci.* **193**, 477–483 (2024).
71. Jiang, Y. et al. Investigating the accuracy of blood oxygen saturation measurements in common consumer smartwatches. *PLOS Digit. Health* **2**, e0000296 (2023).
72. Rajakariar, K. et al. Accuracy of smartwatch pulse oximetry measurements in hospitalized patients with coronavirus disease 2019. *Mayo Clin. Proc. Digit. Health* **2**, 152–158 (2024).
73. Spaccarotella, C. et al. Assessment of non-invasive measurements of oxygen saturation and heart rate with an apple smartwatch: comparison with a standard pulse oximeter. *J. Clin. Med.* **11**, 1467 (2022).
74. Apple. *Blood Oxygen app on Apple Watch*. <https://apps.apple.com> (2022).
75. Rafi, J., Bachman, T. E., Rafi-Huttova, V., Walzel, S. & Rozanek, M. Commercial smartwatch with pulse oximeter detects short-time hypoxemia as well as standard medical-grade device: Validation study. *Digit. health* **8**, 20552076221132127 (2022).
76. Walzel, S. et al. Evaluation of leading smartwatches for the detection of hypoxemia: comparison to reference oximeter. *Sensors* **23**, 9164 (2023).
77. Jiang, Y. et al. Performance of wearable pulse oximetry during controlled hypoxia induction. *medRxiv* <https://doi.org/10.1101/2024.07.16.24310506> (2024).
78. Lee, M., Lee, H. & Park, S. Accuracy of swimming wearable watches for estimating energy expenditure. *Int. J. Appl. Sports Sci.* **30**, 80–90 (2018).
79. Sun, X. et al. Validity of apple watch 6 and Polar A370 for monitoring energy expenditure while resting or performing light to vigorous physical activity. *J. Sci. Med. Sport* **26**, 482–486 (2023).
80. Veerabhadrapa, P. et al. Tracking steps on an Apple Watch at different walking speeds. *J. Gen. Intern. Med.* **33**, 795–796 (2018).
81. Bunn, J. A., Jones, C., Oliveira, A. & Webster, M. J. Assessment of step accuracy using the consumer technology association standard. *J. Sports Sci.* **37**, 244–248 (2019).
82. Benning, N.-H., Knaup, P. & Rupp, R. Measurement performance of activity measurements with newer generation of Apple Watch in wheelchair users with spinal cord injury. *Methods Inf. Med.* **60**, e103–e110 (2021).
83. Glasheen, E., Domingo, A. & Kressler, J. Accuracy of Apple Watch fitness tracker for wheelchair use varies according to movement frequency and task. *Ann. Phys. Rehabil. Med.* **64**, 101382 (2021).
84. Karinharju, K. S. et al. Validity of the Apple Watch® for monitoring push counts in people using manual wheelchairs. *J. Spinal Cord. Med.* **44**, 212–220 (2021).
85. Lambe, R., O'Grady, B., Baldwin, M. & Doherty, C. Investigating the accuracy of apple watch VO2 max measurements: a validation study. *PLoS ONE* **20**, e0323741 (2025).
86. Robbins, R. et al. Accuracy of three commercial wearable devices for sleep tracking in healthy adults. *Sensors (Basel)* **24**, 6532 (2024).
87. Apple. *Estimating sleep stages from Apple Watch*. https://www.apple.com/health/pdf/Estimating_Sleep_Stages_from_Apple_Watch_Oct_2025.pdf (2023).
88. Lee, T. et al. Accuracy of 11 wearable, nearable, and airable consumer sleep trackers: prospective multicenter validation study. *JMIR Mhealth Uhealth* **11**, e50983 (2023).
89. Laukkanen, J. A., Isiozor, N. M. & Kunutsor, S. K. Objectively assessed cardiorespiratory fitness and all-cause mortality risk: an updated meta-analysis of 37 cohort studies involving 2,258,029 participants. *Mayo Clin. Proc.* **97**, 1054–1073 (2022).
90. Koerber, D., Khan, S., Shamsheri, T., Kirubakaran, A. & Mehta, S. Accuracy of heart rate measurement with wrist-worn wearable devices in various skin tones: a systematic review. *J. Racial Ethn. Health Disparities* **10**, 2676–2684 (2023).
91. Byrne, J. et al. Investigating the accuracy of wheelchair push counts measured by fitness watches: a systematic review. *Cureus* **15**, e45322 (2023).

92. Singh, B. et al. Real-world accuracy of wearable activity trackers for detecting medical conditions: systematic review and meta-analysis. *JMIR Mhealth Uhealth* **12**, e56972 (2024).
93. Belani, S., Wahood, W., Hardigan, P., Placzek, A. N. & Ely, S. Accuracy of detecting atrial fibrillation: a systematic review and meta-analysis of wrist-worn wearable technology. *Cureus* **13**, e20362 (2021).
94. Giebel, G. D. & Gissel, C. Accuracy of mhealth devices for atrial fibrillation screening: systematic review. *JMIR Mhealth Uhealth* **7**, e13641 (2019).
95. Nazarian, S., Lam, K., Darzi, A. & Ashrafian, H. Diagnostic accuracy of smartwatches for the detection of cardiac arrhythmia: systematic review and meta-analysis. *J. Med. Internet. Res.* **23**, e28974 (2021).
96. Windisch, P., Schröder, C., Förster, R., Cihoric, N. & Zwahlen, D. R. Accuracy of the apple watch oxygen saturation measurement in adults: a systematic review. *Cureus* **15**, e35355 (2023).
97. Borenstein, M., Hedges, L. V., Higgins, J. P. & Rothstein, H. R. *Introduction to Meta-Analysis* 1st edn, Vol. 452 (John Wiley & Sons, 2021).
98. O'Driscoll, R. et al. How well do activity monitors estimate energy expenditure? A systematic review and meta-analysis of the validity of current technologies. *Br. J. Sports Med.* **54**, 332–340 (2020).
99. Chevanne, G. et al. Accuracy and precision of energy expenditure, heart rate, and steps measured by combined-sensing Fitbits against reference measures: systematic review and meta-analysis. *JMIR mHealth uHealth* **10**, e35626 (2022).
100. Miller, D. J., Sargent, C. & Roach, G. D. A validation of six wearable devices for estimating sleep, heart rate and heart rate variability in healthy adults. *Sensors (Basel)* **22**, 6317 (2022).
101. Doherty, C., Baldwin, M., Lambe, R., Burke, D. & Altini, M. Readiness, recovery, and strain: an evaluation of composite health scores in consumer wearables. *Transl. Exercise Biomed.* **2**, 2(2025).
102. Dunn, J., Coravos, A., Fanarjian, M., Ginsburg, G. S. & Steinhubl, S. R. Remote digital health technologies for improving the care of people with respiratory disorders. *Lancet Digit. Health* **6**, e291–e298 (2024).
103. Kim, H.-G., Cheon, E.-J., Bai, D.-S., Lee, Y. H. & Koo, B.-H. Stress and heart rate variability: a meta-analysis and review of the literature. *Psychiatry Investig.* **15**, 235 (2018).
104. Nayak, S. K. et al. A Review of Methods and Applications for a Heart Rate Variability. *Anal. Algorithms* **16**, 433 (2023).
105. Behrmann, J. et al. Inferring optical tissue properties from photoplethysmography using hybrid amortized inference. *arXiv* <https://doi.org/10.48550/arXiv.2510.02073> (2025).
106. Banach, M. et al. The association between daily step count and all-cause and cardiovascular mortality: a meta-analysis. *Eur. J. Prevent. Cardiol.* **30**, 1975–1985 (2023).
107. US Food and Drug Administration. Pulse Oximeters for Medical Purposes – Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations. <https://www.fda.gov/media/184896/download> (2025).
108. International Organization for Standardization. *Particular Requirements For Basic Safety And Essential Performance Of Pulse Oximeter Equipment*. <https://cdn.standards.itech.ai> (2025).
109. European Union. *Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on Medical Devices*. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32017R0745> (2025).
110. Molina-Garcia, P. et al. Validity of estimating the maximal oxygen consumption by consumer wearables: a systematic review with meta-analysis and expert statement of the interlive network. *Sports Med.* **52**, 1577–1597 (2022).
111. Argent, R. et al. Recommendations for determining the validity of consumer wearables and smartphones for the estimation of energy expenditure: expert statement and checklist of the interlive network. *Sports Med* **52**, 1817–1832 (2022).
112. Mühlen, J. M. et al. Recommendations for determining the validity of consumer wearable heart rate devices: expert statement and checklist of the INTERLIVE Network. *Br. J. Sports Med.* **55**, 767–779 (2021).
113. Johnston, W. et al. Recommendations for determining the validity of consumer wearable and smartphone step count: expert statement and checklist of the INTERLIVE network. *Br. J. Sports Med.* **55**, 780–793 (2021).
114. Suau, Q. et al. Current knowledge about actigraph GT9X link activity monitor accuracy and validity in measuring steps and energy expenditure: a systematic review. *Sensors (Basel)* **24**, 825 (2024).
115. Dreisbach, S., Rhudy, M., Moran, M., Henriquez, B. & Veerabhadrapa, P. Accuracy of apple watch and actigraphs during overground and treadmill walking. *Hum. Mov.* **26**, 83–90 (2025).
116. Dixon, W. G. et al. Charting a course for smartphones and wearables to transform population health research. *J. Med. Internet Res.* **25**, e42449 (2023).
117. Centre, B. H. F. D. S. *Smartphone And Wearable Data In Cardiovascular Research: Understanding The Views Of The Public And Professionals*. <https://zenodo.org/records/10894877> (2024).
118. Hughes, A., Shandhi, M. M. H., Master, H., Dunn, J. & Brittain, E. Wearable devices in cardiovascular medicine. *Circ. Res* **132**, 652–670 (2023).
119. Weiss, A. J. & Jiang, H. J. Overview of clinical conditions with frequent and costly hospital readmissions by payer. *Agency Healthcare Res. Quality* (2021).
120. Burke, R. E. & Coleman, E. A. Interventions to Decrease Hospital Readmissions: Keys for Cost-effectiveness. *JAMA Intern. Med.* **173**, 695–698 (2013).
121. Herrera, C. A. et al. The World Bank–PAHO Lancet regional health Americas commission on primary health care and resilience in Latin America and the Caribbean. *Lancet Reg. Health Am.* **28**, 100643 (2023).
122. Gonçalves-Bradley, D. C. et al. Mobile technologies to support healthcare provider to healthcare provider communication and management of care. *Cochrane Database Syst. Rev.* **8**, Cd012927 (2020).
123. Page, M. J. et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Bmj* **372**, n71 (2021).
124. Taylor, K. S., Mahtani, K. R. & Aronson, J. K. Extracting data from diagnostic test accuracy studies for meta-analysis. *BMJ Evid. Based Med.* **26**, 19–21 (2021).
125. Mokkink, L. B. et al. COSMIN Risk of Bias tool to assess the quality of studies on reliability or measurement error of outcome measurement instruments: a Delphi study. *BMC Med. Res. Methodol.* **20**, 293 (2020).
126. Tipton, E. & Shuster, J. A framework for the meta-analysis of Bland–Altman studies based on a limits of agreement approach. *Stat. Med.* **36**, 3621–3635 (2017).
127. Higgins, J. P., Thompson, S. G. & Spiegelhalter, D. J. A re-evaluation of random-effects meta-analysis. *J. R. Stat. Soc. Ser. A Stat. Soc.* **172**, 137–159 (2009).
128. Bland, J. M. & Altman, D. Statistical methods for assessing agreement between two methods of clinical measurement. *lancet* **327**, 307–310 (1986).
129. Doebler, P., Holling, H. & Sousa-Pinto, B. *Meta-Analysis of Diagnostic Accuracy with mada*. <https://cran.r-project.org/web/packages/mada/vignettes/mada.pdf> (2023).

130. Higgins, J. P., Thompson, S. G., Deeks, J. J. & Altman, D. G. Measuring inconsistency in meta-analyses. *Bmj* **327**, 557–560 (2003).
131. Deeks, J. J., Higgins, J. P., Altman, D. G. & Group, C. S. M. *Analysing Data and Undertaking Meta-Analyses*. <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-10> (2019).
132. Montalvo, S. et al. Commercial smart watches and heart rate monitors: A concurrent validity analysis. *The Journal of Strength & Conditioning Research* **37**, 1802–1808 (2023).
133. Alfonso, C. et al. Agreement between two photoplethysmography-based wearable devices for monitoring heart rate during different physical activity situations: a new analysis methodology. *Scientific reports* **12**, 15448 (2022).
134. Dürking, P. et al. Wrist-worn wearables for monitoring heart rate and energy expenditure while sitting or performing light-to-vigorous physical activity: validation study. *JMIR mHealth and uHealth* **8**, e16716 (2020).
135. Lee, C. & Chow, C. Comparison of Apple watch series 4 vs. Kardiamobile: A tale of two devices. *Canadian Journal of Cardiology* **37**, S43–S44 (2021).
136. Al-Kaisey, A. M. et al. Accuracy of wrist-worn heart rate monitors for rate control assessment in atrial fibrillation. *International journal of cardiology* **300**, 161–164 (2020).
137. Pasadyn, S. R. et al. Accuracy of commercially available heart rate monitors in athletes: a prospective study. *Cardiovascular diagnosis and therapy* **9**, 379 (2019).
138. Hwang, J. et al. Assessing accuracy of wrist-worn wearable devices in measurement of paroxysmal supraventricular tachycardia heart rate. *Korean circulation journal* **49**, 437–445 (2019).
139. Heyken, M. et al. Comparison of wearables for self-monitoring of heart rate in coronary rehabilitation patients. *Georgian medical news* **315**, 78–85 (2021).
140. Huynh, P. et al. Heart rate measurements in patients with obstructive sleep apnea and atrial fibrillation: Prospective pilot study assessing apple watch's agreement with telemetry data. *JMIR cardio* **5**, e18050 (2021).
141. Khushhal, A. et al. Validity and reliability of the Apple Watch for measuring heart rate during exercise. *Sports medicine international open* **1**, E206–E211 (2017).
142. Thomson, E. A. et al. Heart rate measures from the Apple Watch, Fitbit Charge HR 2, and electrocardiogram across different exercise intensities. *Journal of sports sciences* **37**, 1411–1419 (2019).
143. Gillinov, S. et al. Variable accuracy of wearable heart rate monitors during aerobic exercise. *Medicine & Science in Sports & Exercise* **49**, 1697–1703 (2017).
144. Sequeira, N. et al. Common wearable devices demonstrate variable accuracy in measuring heart rate during supraventricular tachycardia. *Heart Rhythm* **17**, 854–859 (2020).
145. Apple. Blood Oxygen app on AppleWatch. https://www.apple.com/healthcare/docs/site/Blood_Oxygen_app_on_Apple_Watch_October_2022.pdf (2022).
146. Apple. Estimating sleep stages from Apple Watch. https://www.apple.com/healthcare/docs/site/Estimating_Sleep_Stages_from_Apple_Watch_Sept_2023.pdf (2023).
147. Apple. Hypertension Notification Feature on Apple Watch. <https://apple.com/health> (2025)..
148. Apple. Using Apple Watch to Estimate Six-Minute Walk Distance. https://www.apple.com/healthcare/docs/site/Using_Apple_Watch_to_Estimate_Six_Minute_Walk_Distance.pdf.
149. Fischer, T. et al. Are smartwatches a suitable tool to monitor noise exposure for publichealth awareness and otoprotection?. *Front Neurol.* **13**, 856219 (2022).

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Author contributions

R.L., C.D., and B.O'G. conceived and designed the study. R.L., C.D., B.O'G., M.S., M.B., and B.C. contributed to the methods of the study. R.L. and B.O'G. conducted the searches of all databases. R.L., M.B., and B.O'G. selected the articles and extracted the data. R.L., M.B., and B.O'G. analysed the data. R.L., M.B., B.O'G., and C.D. accessed and verified the data. R.L. and M.B. assessed risk of bias and R.L. conducted meta-analyses. R.L., M.B., and C.D. wrote the first draft of the manuscript. All authors contributed to data interpretation, revision, and writing of the final version of the manuscript. All authors critically reviewed and approved the content of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Competing interests

The authors declare no competing interests.

Additional information

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